Ray Zirui Zhang

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ACADEMIC POSITIONS

Assistant Professor — Aug 2025 – Present
Department of Mathematical Sciences — Worcester, USA
Worcester Polytechnique Institute

Visiting Assistant Professor — July 2022 – June 2025
Department of Mathematics — Irvine, USA
University of California, Irvine

RESEARCH SUMMARY

My research interests lie in computational mathematics, scientific machine learning, and data science, with applications in biophysics and cancer research.

EDUCATION

Ph.D. in Mathematics — Sept 2017 – June 2022
University of California, San Diego — San Diego, USA

M.S. in Computational Science — Sept 2017 – June 2019
University of California, San Diego — San Diego, USA

B.Eng in Civil Engineering — Sept 2012 – June 2016
University of Hong Kong — Hong Kong

GRANTS & FUNDING

NSF, *Collaborative Research: FDT-BioTech: Mathematical Foundations of Adaptable Digital Twins for Glioblastoma*, $247,931 (PI) — 2027
NVIDIA Academic Grant — 2025

AWARDS & HONORS

Scholar-in-Residence, *AI Long Program for Image-Based Scientific Machine Learning*, National Institute for Theory and Mathematics in Biology (NITMB) — 2027
Keynote speaker for the session *Computational Models and Methods for Predicting Cancer Progression and Treatment Response* at WCCM 2024 — 2024
UCI Systems Biology Opportunity Award (with Rany Tith and Song Zhu) — 2024
Rising Star speaker at UCI Center for Complex Biological Systems Retreat — 2024
The Data Science Conference 2023 travel grant (NSF) — 2023
Powell Dissertation Award (UCSD) — 2022
Gold Medal of BirdCLEF 2021 – Birdcall Identification (Kaggle) — 2021
James B. Ax Fellowship (UCSD) — 2019
Emerging Engineering Award (Institute of Civil Engineers) — 2017
Soong Ching Ling Scholarship (Soong Ching Ling Foundation) — 2012

PUBLICATIONS & PREPRINTS

BiLO: Bilevel Local Operator Learning for PDE Inverse Problems
Ray Zirui Zhang, Christopher Miles, Xiaohui Xie, John Lowengrub
Journal of Computational Physics, 2026

Bayesian BiLO: Bilevel Local Operator Learning for Efficient Uncertainty Quantification of Bayesian PDE Inverse Problems with Low-Rank Adaptation
Ray Zirui Zhang, Christopher Miles, Xiaohui Xie, John Lowengrub
Accepted, Nature Communications, 2026

Identifiability Limits of Physics-Informed Inference for Spatial Stochastic Dynamics from Static Snapshots
Rujie Gu, Ray Zirui Zhang, Christopher E. Miles
Preprint on arXiv, 2026

Personalized Predictions of Glioblastoma Infiltration: Mathematical Models, Physics-Informed Neural Networks and Multimodal Scans
Ray Zirui Zhang, Ivan Ezhov, Michal Balcerak, Andy Zhu, Benedikt Wiestler, Bjoern Menze, John Lowengrub
Medical Image Analysis, 2025

Individualizing Glioma Radiotherapy Planning by Optimization of a Data and Physics Informed Discrete Loss
Michal Balcerak, Ivan Ezhov, Petr Karnakov, Sergey Litvinov, Petros Koumoutsakos, Jonas Weidner, Ray Zirui Zhang, John S Lowengrub, Bene Wiestler, Bjoern Menze
Nature Communications, 2025

A Compact Coupling Interface Method with Second-Order Gradient Approximation for Elliptic Interface Problems
Ray Zirui Zhang, Li-Tien Cheng
Journal of Scientific Computing, 2024

Binary Level Set method for Variational Implicit Solvation Model
Ray Zirui Zhang, Li-Tien Cheng
SIAM Journal of Scientific Computing, 2023

Explicit-Solute Implicit-Solvent molecular simulation with Binary level-set, adaptive-mobility, and GPU
Shuang Liu, Ray Zirui Zhang, Li-Tien Cheng, and Bo Li
Journal of Computational Physics, 2022

Coupling Monte Carlo, Variational Implicit Solvation, and Binary Level-Set for Simulations of Biomolecular Binding
Ray Zirui Zhang, Clarisse G. Ricci, Chao Fan, Li-Tien Cheng, Bo Li, J. Andrew McCammon
Journal of Chemical Theory and Computation, 2021

Runout Scaling and Deposit Morphology of Rapid Mudflows
Lu Jing, C. Y. Kwok, Y. F. Leung, Ray Zirui Zhang, Lutao Dai
Journal of Geophysical Research: Earth Surface, 2018

TALKS

Applied Mathematics and Computation Seminar, UMass Amherst (2026)
Applied and Interdisciplinary Mathematics Seminar, Northeastern University (2026)
Interdisciplinary Center for Quantitative Modeling in Biology, UCR (2025)
The Mathematics of Data, Dynamics, and Life Sciences, UCI (2025)
Center for Computational Mathematics Seminar, UCSD (2025)
Computational and Applied Mathematics Seminar, Arizona State University (2025)
AMS Fall Western Sectional Meeting, UCR (2024)
Computational and Applied Mathematics Colloquium, Penn State (2024)
16th World Congress on Computational Mechanics (WCCM) and 4th Pan American Congress on Computational Mechanics, Vancouver, Canada (2024)
Joint Annual Meeting of the Korean Society for Mathematical Biology and the Society for Mathematical Biology (SMB), Seoul, Korea (2024)
SIAM Conference on Life Science, Portland, OR (2024)
2024 UCI Systems Biology Retreat, Pasadena, CA (2024)
Statistical and Machine Learning Applications in Biomedical Sciences, UCI (2024)
Center for Computational Mathematics Seminar, UCSD (2024)
17th U.S. National Congress on Computational Mechanics, Albuquerque, NM (2023)
The Data Science Conference, University of San Francisco (2023)
Southern California Applied Mathematics Symposium (2023)
UCI Applied Math Seminar (2023)
SIAM Conference on Optimization, Seattle, WA (2023)
Long Feng Science Forum, The Chinese University of Hong Kong, Shenzhen (2023)
Predictive Modeling in Biology and Medicine Conference, UC Riverside (2023)
The International Conference on New Trends in Computational and Data Sciences, Caltech (Poster, 2023)
SIAM Texas Louisiana Section Annual Meeting, University of Houston (2022)

TEACHING

Instructor, WPI Aug 2025 – Present
MA 1021 Calculus I (A2026)
MA 510 Numerical Methods (Fall2026)
MA 2621 Probability for Applications (A2025)
MA 1024 Calculus IV (C2026)
MA 2631 Probability Theory (D2026)

Instructor, UCI Sept 2022 – July 2025
MATH 9 Intro Programming for Numerical Analysis (FA22, SP23, SP24)
MATH 10 Intro Programming for Data Science (SP24, FA24, SP25)
MATH 199 Undergraduate Supervised Reading and Research

Teaching Assistant, UCSD Sept 2018 – Jun 2022

MATH 114 Intro Computational Stochastics
MATH 170A Intro Numerical Analysis: Linear Algebra
MATH 170B Intro Numerical Analysis: Approximation and Nonlinear Equations
MATH 170C Intro Numerical Analysis: Ordinary Differential Equations
MATH 174 Numerical Methods and Physical Modeling
MATH 180A Intro Probability
MATH 181A Intro Statistics
MATH 270A Numerical Linear Algebra
MATH 270C Numerical Ordinary Differential Equations

SERVICES

Mentor of Math ExpLR: Summer Research Program for High School Students	2024
Organizer of Minisymposium *Image-Guided Mathematical Modeling in Clinical and Pre-Clinical Oncology* at SMB annual meeting	2024
Guest Lecturer of UCI Cancer Systems Biology Short Course	2024
Reviewer of NeurIPS, ICML, ICLR, TMLR	
Organizer of 2023 Southern California Applied Mathematics Symposium	2023